Racing-grade hydrodynamics for uncrewed maritime systems

Wyspa Technologies is a small US engineering firm that predicts, designs and builds fast boats for Grand Prix sailing. We apply our in-house hydrofoil, hull-performance and composite-structures tools to small, fast and efficient uncrewed surface vessels.

ENTITY	UEI	CAGE	BUSINESS TYPE
Wyspa Technologies LLC	NWJRCE69AWH9	Pending	Small business

Core competencies

- **Velocity & hydrofoil performance prediction.** In-house VPP with Delft-series resistance, ITTC-57 friction, lifting-line foils, and ventilation and cavitation limits. Validated against measured TP52 and Swan 42 data.
- **Composite laminate engineering.** Classical laminate theory with progressive ply failure (Tsai-Hill, Tsai-Wu), 550+ reinforcements and 90+ cores, and 3D-to-2D ply flattening with drape prediction and DXF cut files.
- **Foil flight dynamics & control.** Time-domain simulation of foil-borne craft in irregular seas with closed-loop ride-height, pitch and roll control.
- **Survey, scanning & compliance.** FARO 3D scanning and as-built models, stability analysis and USCG compliance documentation, CAD and cut files for builders.

Lead concept: autonomous foiling USV

GENERIC 7 M, 1,500 KG ELECTRIC USV	FOIL-BORNE	PLANING
Energy at 23 kn, calm	1.13 kWh/nm	2.39 kWh/nm
RMS vertical accel. SS2 / SS3 / SS4	0.06/0.10/0.28 g	0.56/0.91/0.93 g

Simulation estimates from Wyspa's time-domain model; not yet validated by tank or sea trials.

Differentiators

- Engineering proven in Grand Prix sailing, where small speed margins decide results.
- We own all core tools, so we can change them without licensing barriers and cycle quickly from analysis to design.

Past performance (commercial)

- 90+ yacht-performance and engineering projects, including TP52, Swan 42 and Etchells programs: VPP, rating-rule optimization, race-data analysis and weather routing.
- Structural engineering for a commercial electric-aircraft program.

NAICS

541330 · 541370 · 541380 · 541690 · 541715

Applications

- Harbor and port security · distributed logistics · ocean sensing
- Hydrodynamic and composite subcontracts for USV builders

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